# Chengyi Zhou

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## Profile

Master's student in Computer and Systems Sciences at Stockholm University. For my Master's thesis, in collaboration with KTH Royal Institute of Technology, I work on reward hacking in reinforcement learning for language models. Background in Android performance and stability, GPU acceleration, and data-driven monitoring. Interested in language model alignment and reliable evaluation of LLM-based agents.

## Education

**Stockholm University** **Stockholm, Sweden**
*MSc in Computer and Systems Sciences* *2025–2027*

- Master's thesis: reward hacking in reinforcement learning for language models, in collaboration with Martin Monperrus's group at KTH.
- Supervisor: Nicolas Harrand (Stockholm University); co-supervisor: André Silva (KTH).

**East China Normal University** **Shanghai, China**
*BEng in Computer Science and Technology* *2020–2024*

## Experience

**Kuaishou** **Hangzhou, China**
*Android Engineer* *Jul 2024–Jul 2025*

- Developed an interaction-heatmap SDK that visualised in-app usage patterns, helping product teams identify UI and usability issues.
- Investigated and helped address Android Application Not Responding (ANR) incidents by analysing reported stack traces and execution behaviour.
- Built and maintained BI dashboards for performance and operational metrics, enabling team-level monitoring and data-informed decisions.

**Danzhu Sports Vision** **Shanghai, China**
*Algorithm Engineering Intern* *Mar 2024–Jun 2024*

- Migrated a motion-analysis engine from CPU-based processing to GPU rendering with OpenGL and CUDA.
- Integrated NVIDIA NVENC hardware encoding to improve video-output efficiency while retaining high visual quality.

## Projects

**2D Game Engine**

- Built a modular 2D game engine in C++ using OpenGL, ImGui, and Box2D, with real-time rendering, physics simulation, and UI integration.
- Designed a developer-friendly architecture to support rapid prototyping of interactive games.

**3D Model Slicing System**

- Developed a WebGL-based system that generates real-time sectional views and G-code for 3D-printing workflows.
- Implemented an interactive interface for uploading, manipulating, and slicing 3D models.

**Livestock Monitoring Android Application**

- Developed a production-ready Android application for real-time livestock monitoring using Kotlin, Flow, and MVVM architecture.
- Integrated IoT telemetry via REST APIs for live status updates and automated alerts; delivered critical-event notifications with Firebase Cloud Messaging.
- Built Material Design 3 dashboards with dynamic charts for monitoring and data visualisation.

## Technical Skills

**Programming**: C, C++, Python, Kotlin, Java, SQL

**Systems & graphics**: Android, OpenGL, CUDA, WebGL

**Data & monitoring**: Hive, ClickHouse, ETL pipelines, BI dashboards